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Project Report – Tax Law Question Answering System

1. Setup

Inference task:

For the inference task the model the llama-4-scout-17b-16e-instruct via Groq Cloud was used. The Hyperparameters included are a temperature of 0.1 and max tokens of 150. The model uses top-p sampling as standard approach. Furthermore, the model was prompted with a system instruction to provide factual and accurate answers based on the query. No specific legal context or examples were provided to establish a pure baseline for comparison. The model was pre-trained on a mix of publicly available, licensed data and information from Meta's products and services. This includes publicly shared posts from Instagram and Facebook and people's interactions with Meta AI according to huggingface.

Fine-Tuning task:

For the training the model Mistral-7B-v0.3 was used and a dataset with new questions and answers including the laws which were used in the shared dataset was created as a training and validation set. As method a QLoRA (4-bit Quantization) was used in order to compress the model to use it on Kaggle's GPU. The LoRA Hyperparameters include a rank of 16, alpha 32 to adapt the model to Austrian tax terminology. The training targeted the attention layers to increase the models understanding of law. For the training parameters I used 3 epochs, batch size 2 (as the VRAM is very limited on Kaggle) and 8 accumulation steps to keep the training stable and to simulate a batch size of 16.

RAG task:

For the RAG task paraphrase-multilingual-MiniLM-L12-v2 was used as the retrieval model for generating embeddings. The model was chosen, because the data processed by the model is in German and therefore a multilingual model was used. For the vector database FAISS (Facebook AI Similarity Search) was used to search for similarities between the prompt and the documents provided to the model. 11 Austrian laws were provided to the model for example KStG or EStG in current version. For preprocessing recursive character text splitter with chunk size 800 and chunk overlap of 100 characters was applied. For each query the model searched for the top 2 (k=2) passages as context for the generation.

2. Evaluation Results

To evaluate performance the dataset created in the shard task with the columns ID and correct_answer was used as reference. In order to ensure a fair comparison across all models the following three metrics are used:

- BERTScore (F1) which measures the semantic similarity by comparing the meaning of words using contextual embeddings rather than just exact word matches.
- ROUGE-L which measures the structural overlap by identifying the longest common subsequence of words between the model's response and the reference.
- Cosine Similarity which measures the thematic distance between the two texts, showing how closely the overall topics align.

Result Table

Model	BERTScore (F1)	ROUGE-L	Cosine Similarity
Inference (Llama 4)	**0.7262**	**0.1935**	**0.6510**
Fine-tuning (Mistral-7B-v0.3)	0.7177	0.1590	0.5823
RAG (Mistral-7B)	0.6828	0.1356	0.5862

From a statistical perspective the inference model performed best. This is primarily because it generated shorter responses in a more general tone. In contrast, the other models adopted a more specialized juristic tone, leading to a lower structural match with the reference dataset.

3. Error Analysis

While evaluating the models three types of errors across the models were identified:

1. The fine-tuned model successfully adopted the Austrian law vocabulary. However, it occasionally suffered from citation mixing, for example correctly explaining a tax concept but citation of EStG instead of the KStG. This suggests that while semantic concepts are well-learned, the exact mapping of sources is challenging for the model.
2. The RAG model occasionally suffers from distraction. If a retrieved document chunk contained secondary legal notes, the model sometimes included this irrelevant information. Furthermore, the model includes conversational fillers like "Based on the provided text", which negatively impacted the overlap-based evaluation metrics like ROUGE-L.
3. A general challenge during the evaluation is that metrics like BERTScore and ROUGE-L penalize longer, more detailed answers. Since the reference dataset contained shorter, less technical answers, the more comprehensive juristic responses from the fine-tuned and RAG model received lower scores despite their factual accuracy.

4. Conclusion

While the inference model achieved the highest statistical scores, this project demonstrates that automated metrics do not always reflect the practical utility. The RAG system provided the most grounded and verifiable answers by directly citing the provided laws, which is crucial for the reliability required in tax law applications.

Future improvements could focus on refining the RAG prompt to reduce source explanations and exploring larger models for fine-tuning to better map legal citations.